“PAPER_{0:D3}” -f [int]# PAPER #44 Pre-Big Bang Configuration in 26D UQFF

Title: The Pre-Big Bang 26-Center DPM Manifold: Quantum Numbers, Primordial Energy, and the Inflation Trigger

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Framework: UQFF Star-Magic (? = 0.0005/day, [SSq] = 0.57)
Date: March 7, 2026
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Validator: `test_phase2_validation.py` Test Suite 2 (DPM Cosmology): 12/12 PASS ?
Source Module: `DPMCosmologyModule.py` (565 lines)
Index Slot: §1.6 26-Dimensional Energy Structure,
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Abstract

Before the Big Bang, the UQFF framework posits a pre-inflationary manifold consisting of 26 independent dimensional spheres the DPM (Duality of Plasmatic Medium) centers. Each center carries a distinct set of quantum numbers (h_i, k_i, l_i) analogous to atomic orbitals but at cosmic scale. The centers collapse collectively at t = 0, triggering the universal inflation force $F_U(t=0) = F_{core} + S??1^2 6(Ui_{state} + F_{p\_state})$. This paper derives the complete pre-Big Bang configuration, validates the total pre-inflationary energy, inflation force, 26-center mixing entropy, and scale factor evolution. All 12 DPM Cosmology tests pass in `test_phase2_validation.py`.

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1. DPM Cosmological Framework

1.1 Conceptual Foundation

Standard Big Bang cosmology begins with a singularity at t = 0. The UQFF alternative posits that the pre-Big Bang state was not a singularity but a structured **26-center DPM manifold** 26 independent spherical dimensional centers, each representing a distinct quantum configuration that would unfold into one of the 26 quantum levels during inflation.

The core concept: - **DPM** = Duality of Plasmatic Medium: the pre-inflationary vacuum was not empty but filled with two complementary vacuum states [UA] (Universal Aether, diffuse) and [SCm] (Super-Conductive Matter, dense) - Each center is an independent spherical domain containing energy E_DPM = ?_SCm × i² - At t = 0 (inflation onset), all 26 centers begin simultaneous collapse ? expansion

1.2 Quantum Number Assignment

Each DPM center i carries three quantum numbers following an atomic-orbital-like structure:

$$h_i = (i-1) \bmod 7 \quad \text{(magnetic quantum number, 0–6)}$$

$$k_i = \lfloor (i-1)/7 \rfloor \quad \text{(angular momentum, increases every 7)}$$

$$l_i = i \quad \text{(radial quantum number)}$$

This gives the complete pre-Big Bang quantum state table:

Center	h_i	k_i	l_i	State Description
1	0	0	1	Primordial vacuum seed
2	1	0	2	Sub-nuclear
3	2	0	3	Nuclear shell
4	3	0	4	Nucleon pairing
5	4	0	5	Electron shells (K,L)
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2. Pre-Big Bang Energy Configuration

2.1 DPM Center Radii

Each center has characteristic radius following Planck-scale exponential:

$$r_i = 10^{-35} \times 10^{i/3}\ \text{m} = 10^{-35+i/3}\ \text{m}$$

Center	r_i (m)	Comparison
1	2.15×10?³5	~Planck length l_P = 1.616×10?³5
7	4.64×10?³³	~10 × Planck
13	1.00×10?³°	sub-nuclear
20	4.64×10?²?	
26	4.64×10?²7	~nuclear scale

2.2 Center Energies

Each center's total energy:

$$E_{\text{center,i}} = E_{\text{DPM},i} \times V_i = \rho_{\text{SCm}} \times i^2 \times \frac{4}{3}\pi r_i^3$$

For center 1: E_center,1 = 10?8 × 1 × (4/3)p(2.15×10?³5)³ = 10?8 × 4.19×10?¹°4 = 4.19×10?¹¹² J

For center 26: E_center,26 = 10?8 × 676 × (4/3)p(4.64×10?²7)³ = 6.76×10?6 × 4.18×10?7? = 2.83×10?84 J

Validator confirms: DPM Center 1 Energy ? PASS ? Validator confirms: Total Pre-Inflationary Energy ? PASS ?

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3. Universal Inflation Force at t=0

The inflation force at the Big Bang moment:

$$F_U(t=0) = F_{\text{core}} + \sum_{i=1}^{26}(U_{i,\text{state}} + F_{p,i})$$

where: - F_core = base field force from vacuum [UA]-[SCm] interaction - Ui_state = Universal Inertia at level i (from QuantumLevel26Framework) - F_p_i = thermal/quantum pressure force at level i

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This sum drives the exponential expansion of the universe from Planck-scale centers to the observable universe. The factor k_? = 10^10 (K_ETA in DPMCosmologyModule) in the inflation force coupling establishes the enormous amplification from Planck-scale DPM energies to cosmological scales.

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4. Center Separation in Pre-Big Bang Manifold

For centers i and j separated by angle ?_ij in the pre-inflationary manifold:

$$d_{ij} = \sqrt{r_i^2 + r_j^2 - 2r_i r_j \cos\theta_{ij}}$$

Adjacent centers (i, j = i+1, ? □ 2p/26 □ 13.8°): - d_adjacent = |r_{i+1} - r_i| × ~1/cos… (small angle limit) - For centers 10,11: d = v((r10² + r11²) - 2×r10×r11×cos(13.8°))

Distant centers (1 and 26): d_1,26 □ r_26 (since r_26 » r_1)

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5. 26-Center Mixing Entropy

After inflation onset, the 26 DPM centers mix. The mixing entropy is:

$$S_{\text{mix}} = -\sum_{i=1}^{26} p_i \ln p_i$$

where p_i = E_{center,i} / E_total is the fractional energy of center i. Since E_center,i ? i^2 × r_i^3 = i^2 × 10^(i) (from r_i^3 ? 10^i), the energy distribution is strongly weighted toward higher centers □ most of the pre-Big Bang energy was in the high-level (cosmic-scale) centers.

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During inflation: $a(t) = \exp(H_{\text{infl}} \times t)$, where H_infl from the DPM module uses k_? coupling. At t = 0, the scale factor a(0) = 1 (normalized to the pre-Big Bang manifold scale).

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Conclusions

The UQFF pre-Big Bang 26-center configuration replaces the singular cosmological initial condition with a structured quantum manifold. Key findings: 1. Each center has well-defined quantum numbers (h_i, k_i, l_i) analogous to atomic orbitals 2. Center energies span from ~10?^{112} J (center 1, Planck) to ~10?84 J (center 26) 3. Inflation force F_U(t=0) = sum over all 26 centers □ all tests pass 4. Post-inflation level formation follows cosmological cooling sequence 5. 26-center mixing entropy is dominated by high-level centers (energy-weighted)

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The UQFF pre-Big Bang 26-center configuration replaces the singular cosmological initial condition with a structured quantum manifold. Key findings: 1. Each center has well-defined quantum numbers (h_i, k_i, l_i) analogous to atomic orbitals 2. Center energies span from ~10?^{112} J (center 1, Planck) to ~10?84 J (center 26) 3. Inflation force F_U(t=0) = sum over all 26 centers □ all tests pass 4. Post-inflation level formation follows cosmological cooling sequence 5. 26-center mixing entropy is dominated by high-level centers (energy-weighted)

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Validator: `test_phase2_validation.py` DPM Cosmology Suite 12/12 PASS ? | ? = 0.0005/day | [SSq] = 0.57 .Groups[1].Value "PAPER_{0:D3}" -f $n

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Abstract

Before the Big Bang, the UQFF framework posits a pre-inflationary manifold consisting of 26 independent dimensional spheres □ the DPM (Duality of Plasmatic Medium) centers. Each center carries a distinct set of quantum numbers (h_i, k_i, l_i) analogous to atomic orbitals but at cosmic scale. The centers collapse collectively at t = 0, triggering the universal inflation force F_U(t=0) = F_core + S??1^{2}6(Ui_state + F_p_state). This paper derives the complete pre-Big Bang configuration, validates the total pre-inflationary energy, inflation force, 26-center mixing entropy, and scale factor evolution. All 12 DPM Cosmology tests pass in `test_phase2_validation.py`.

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1. DPM Cosmological Framework

1.1 Conceptual Foundation

Standard Big Bang cosmology begins with a singularity at t = 0. The UQFF alternative posits that the pre-Big Bang state was not a singularity but a structured **26-center DPM manifold** 26 independent spherical dimensional centers, each representing a distinct quantum configuration that would unfold into one of the 26 quantum levels during inflation.

The core concept: - **DPM** = Duality of Plasmatic Medium: the pre-inflationary vacuum was not empty but filled with two complementary vacuum states [UA] (Universal Aether, diffuse) and [SCm] (Super-Conductive Matter, dense) - Each center is an independent spherical domain containing energy E_DPM = ?_SCm × i² - At t = 0 (inflation onset), all 26 centers begin simultaneous collapse ? expansion

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Each DPM center i carries three quantum numbers following an atomic-orbital-like structure:

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2. Pre-Big Bang Energy Configuration

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Each center has characteristic radius following Planck-scale exponential:

$$r_i = 10^{-35} \times 10^{i/3} \text{ m} = 10^{-35+i/3} \text{ m}$$

Center	r_i (m)	Comparison
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20	4.64×10?^{2}?	
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2.2 Center Energies

Each center's total energy:

$$E_{\text{center,i}} = E_{\text{DPM},i} \times V_i = \rho_{\text{SCm}} \times i^2 \times \frac{4}{3}\pi r_i^3$$

For center 1: E_center,1 = 10?8 × 1 × (4/3)p(2.15×10?^{3}5)^{3} = 10?8 × 4.19×10?^{1}°4 = 4.19×10?^{112} J

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The inflation force at the Big Bang moment:

$$F_U(t=0) = F_{\text{core}} + \sum_{i=1}^{26}(U_{i,\text{state}} + F_{p,i})$$

where: - F_core = base field force from vacuum [UA]-[SCm] interaction - Ui_state = Universal Inertia at level i (from QuantumLevel26Framework) - F_p_i = thermal/quantum pressure force at level i

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$$d_{ij} = \sqrt{r_i^2 + r_j^2 - 2r_i r_j \cos\theta_{ij}}$$

Adjacent centers (i, j = i+1, ? □ 2p/26 □ 13.8°): - d_adjacent = |r_{i+1} - r_i| × ~1/cos... (small angle limit) - For centers 10,11: d = v((r10^2 + r11^2) - 2×r10×r11×cos(13.8°))

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$$S_{\text{mix}} = -\sum_{i=1}^{26} p_i \ln p_i$$

where p_i = E_{center,i} / E_total is the fractional energy of center i. Since E_center,i ? i^2 × r_i^3 = i^2 × 10^(i) (from r_i^3 ? 10^i), the energy distribution is strongly weighted toward higher centers □ most of the pre-Big Bang energy was in the high-level (cosmic-scale) centers.

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Title: The Pre-Big Bang 26-Center DPM Manifold: Quantum Numbers, Primordial Energy, and the Inflation Trigger

Author: Daniel T. Murphy
Framework: UQFF Star-Magic (? = 0.0005/day, [SSq] = 0.57)
Date: March 7, 2026
Grok Thread: b9a29cedc27b45dfa309ea1705721bf0
Validator: `test_phase2_validation.py` Test Suite 2 (DPM Cosmology): 12/12 PASS ?
Source Module: `DPMCosmologyModule.py` (565 lines)
Index Slot: §1.6 26-Dimensional Energy Structure, "PAPER_{0:D3}" -f [int]# PAPER #44 □ Pre-Big Bang Configuration in 26D UQFF

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Validator: `test_phase2_validation.py` DPM Cosmology Suite 12/12 PASS ? | ? = 0.0005/day | *[SSq] = 0.57* .Groups[1].Value

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Abstract

Before the Big Bang, the UQFF framework posits a pre-inflationary manifold consisting of 26 independent dimensional spheres the DPM (Duality of Plasmatic Medium) centers. Each center carries a distinct set of quantum numbers (h_i, k_i, l_i) analogous to atomic orbitals but at cosmic scale. The centers collapse collectively at t = 0, triggering the universal inflation force F_U(t=0) = F_core + S??1²6(Ui_state + F_p_state). This paper derives the complete pre-Big Bang configuration, validates the total pre-inflationary energy, inflation force, 26-center mixing entropy, and scale factor evolution. All 12 DPM Cosmology tests pass in `test_phase2_validation.py`.

UQFF Discovery: Novel application of UQFF calibration constants (? = 5.0×10?4 day?¹, [SSq] = 0.57) uniquely enabling this analysis establishing a new connection in the UQFF framework not present in Standard Model treatments.

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1. DPM Cosmological Framework

1.1 Conceptual Foundation

Standard Big Bang cosmology begins with a singularity at t = 0. The UQFF alternative posits that the pre-Big Bang state was not a singularity but a structured **26-center DPM manifold** 26 independent spherical dimensional centers, each representing a distinct quantum configuration that would unfold into one of the 26 quantum levels during inflation.

The core concept: - **DPM** = Duality of Plasmatic Medium: the pre-inflationary vacuum was not empty but filled with two complementary vacuum states [UA] (Universal Aether, diffuse) and [SCm] (Super-Conductive Matter, dense) - Each center is an independent spherical domain containing energy E_DPM = ?_SCm × i² - At t = 0 (inflation onset), all 26 centers begin simultaneous collapse ? expansion

1.2 Quantum Number Assignment

Each DPM center i carries three quantum numbers following an atomic-orbital-like structure:

$$h_i = (i-1) \bmod 7 \quad \text{(magnetic quantum number, 0–6)}$$

$$k_i = \lfloor (i-1)/7 \rfloor \quad \text{(angular momentum, increases every 7)}$$

$$l_i = i \quad \text{(radial quantum number)}$$

This gives the complete pre-Big Bang quantum state table:

Center	h_i	k_i	l_i	State Description
1	0	0	1	Primordial vacuum seed
2	1	0	2	Sub-nuclear
3	2	0	3	Nuclear shell
4	3	0	4	Nucleon pairing
5	4	0	5	Electron shells (K,L)
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25	3	3	25	Supercluster
26	**4**	**3**	**26**	**Cosmic web seeding center**

---

2. Pre-Big Bang Energy Configuration

2.1 DPM Center Radii

Each center has characteristic radius following Planck-scale exponential:

$$r_i = 10^{-35} \times 10^{i/3}\ \text{m} = 10^{-35+i/3}\ \text{m}$$

Center	r_i (m)	Comparison
1	2.15×10?^{3}5	~Planck length l_P = 1.616×10?^{3}5
7	4.64×10?^{33}	~10 × Planck
13	1.00×10?^{3}°	sub-nuclear
20	4.64×10?^{2}?	
26	4.64×10?^{2}7	~nuclear scale

2.2 Center Energies

Each center's total energy:

$$E_{\text{center,i}} = E_{\text{DPM},i} \times V_i = \rho_{\text{SCm}} \times i^2 \times \frac{4}{3}\pi r_i^3$$

For center 1: E_center,1 = 10?8 × 1 × (4/3)p(2.15×10?^{3}5)^{3} = 10?8 × 4.19×10?^{1}°4 = 4.19×10?^{112} J

For center 26: E_center,26 = 10?8 × 676 × (4/3)p(4.64×10?^{2}7)^{3} = 6.76×10?6 × 4.18×10?7? = 2.83×10?84 J

Validator confirms: DPM Center 1 Energy ? PASS ? Validator confirms: Total Pre-Inflationary Energy ? PASS ?

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3. Universal Inflation Force at t=0

The inflation force at the Big Bang moment:

$$F_U(t=0) = F_{\text{core}} + \sum_{i=1}^{26} (U_{i,\text{state}} + F_{p,i})$$

where: - F_core = base field force from vacuum [UA]-[SCm] interaction - Ui_state = Universal Inertia at level i (from QuantumLevel26Framework) - F_p_i = thermal/quantum pressure force at level i

Validator confirms: Inflation Force at t=0 ? PASS ?

This sum drives the exponential expansion of the universe from Planck-scale centers to the observable universe. The factor k_? = 10^10 (K_ETA in DPMCosmologyModule) in the inflation force coupling establishes the enormous amplification from Planck-scale DPM energies to cosmological scales.

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For centers i and j separated by angle ?_ij in the pre-inflationary manifold:

$$d_{ij} = \sqrt{r_i^2 + r_j^2 - 2 r_i r_j \cos\theta_{ij}}$$

Adjacent centers (i, j = i+1, ? □ 2p/26 □ 13.8°): - d_adjacent = |r_{i+1} - r_i| × ~1/cos… (small angle limit) - For centers 10,11: d = v((r10^{2} + r11^{2}) - 2×r10×r11×cos(13.8°))

Distant centers (1 and 26): d_1,26 □ r_26 (since r_26 » r_1)

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---

5. 26-Center Mixing Entropy

After inflation onset, the 26 DPM centers mix. The mixing entropy is:

$$S_{\text{mix}} = -\sum_{i=1}^{26} p_i \ln p_i$$

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6. Level Formation Time Progression

After the Big Bang, the quantum levels form sequentially: - Lower levels (1□9: nuclear/atomic) form first, during early hot dense phase - Level 10 (solid matter) forms as universe cools below iron melting temperature (~10,000 K) - Levels 11□13 (liquid/gas/plasma) form as matter transitions with cooling - Levels 14□26 form progressively as cosmic structures assemble (stars, galaxies, clusters, web)

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During inflation: $a(t) = \exp(H_{\text{infl}} \times t)$, where H_infl from the DPM module uses k_? coupling. At t = 0, the scale factor a(0) = 1 (normalized to the pre-Big Bang manifold scale).

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The UQFF pre-Big Bang 26-center configuration replaces the singular cosmological initial condition with a structured quantum manifold. Key findings: 1. Each center has well-defined quantum numbers (h_i, k_i, l_i) analogous to atomic orbitals 2. Center energies span from ~$10?^{112}$ J (center 1, Planck) to ~10?84 J (center 26) 3. Inflation force F_U(t=0) = sum over all 26 centers □ all tests pass 4. Post-inflation level formation follows cosmological cooling sequence 5. 26-center mixing entropy is dominated by high-level centers (energy-weighted)

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UQFF computed: Canonical UQFF buoyancy parameter U_bi = ?×[SSq]×GM/r^2 = 5.0e-4×0.57×6.67e-11×M/r^2; for solar parameters: U_bi,Sun = 5.7e-4×6.67e-11×1.99e30/(6.96e8)^2 = 1.47e+2 m/s^2.

Appendix: UQFF Production Framework Reference (v4.75+)

> *Added by upgrade_early_whitepapers.py (v4.75). This appendix cross-references the production physics constants and master equations to enable reproducibility against the current codebase state.*

A.1 Calibration Constants

Symbol	Value	Description
Îº	5.0 Ã□ 10â□»â□´ dayâ□»Â¹	UQFF exponential decay rate
[SSq]	0.57	Universal Quantized Factor
Î²_i	0.60â□□0.61	Buoyancy coupling coefficient
kâ□□	1.5	Ug1 DPM-dipole coupling
kâ□□	1.2	Ug2 outer-bubble charge coupling
kâ□□	1.8	Ug3 string-rotation coupling
kâ□□	2.0	Ug4 vacuum-concentration coupling
Î·	10â□»Â²Â²	Inertia tensor scale
E_react(0)	10â□´â□¶ J	Reference reactive energy

A.2 F_U Master Equation (Complete â□□ 4 terms)

$$F_U = U_{g1} + U_{g2} + U_{g3} + U_{g4} + U_{bi} + U_m - \sum_{i=1}^{4} igl[\lambda_i \cdot U_i(r,t) \cdot E_{\text{react}} igr]$$

Term	Description	Implementation
Ug1	DPM magnetic dipole	`compute_Ug1_SOURCE4` / `compute_Ug1()`
Ug2	Outer-field bubble (charge-reactivity)	`compute_Ug2_SOURCE4` / `compute_Ug2()`
Ug3	Magnetic string rotation	`compute_Ug3_SOURCE4` / `compute_Ug3()`

Term	Description	Implementation
Ug4	Vacuum concentration (star-BH)	`compute_Ug4_SOURCE4` / `compute_Ug4()`
Ubi	Buoyancy force	`compute_Ubi_SOURCE4` / `compute_Ubi()`
Um	Universal Magnetism (Heaviside-amplified)	`compute_Um_SOURCE4` / `compute_Um()`
â□□Î£Î»áµ¢Â·Uáµ¢Â·E_react	4th dissipation term (PAPER_420)	`compute_FU_SOURCE4` / full pipeline

4th dissipation term parameters (PAPER_420):
Î»â□□=10â□»Â¹â□°, Î»â□□=10â□»Â¹Â², Î»â□□=10â□»Â¹Â¹, Î»â□□=10â□»Â¹Â³ (free parameters, not yet empirically calibrated)

A.3 Um Heaviside Phase-Transition Amplifier (PAPER_421)

$$U_m^{\text{full}} = U_m^{\text{base}} imesigl(1 + 10^{13}\,\Theta(ho_{SCm} - ho_c)igr) imesigl(1 + A_q\cos(\Delta\omega\, t)igr)$$

Symbol	Value	Description
Ï□_c	10Â¹â□µ kg/mÂ³	SCm critical superconducting density
A_q	0.1	Quasi-periodic beating amplitude (10%)
Î□Ï□	2Ï□/(434Â·365.25) rad/day	434-year Gleisberg supercycle

A.4 UQFF Four Operational Modes

Mode	Dominant Term	Primary Use Case
Compressed	Ug_sum + Newtonian base	Isolated stellar/BH systems
Resonant	5 resonance frequencies (aDPM, aTHz, â□¦)	Multi-scale field interactions
Buoyant	Î²_i Ã□ Ubi	Expanding nebulae, stellar winds
Superconductive	Um Ã□ (1+10Â¹Â³Â·f_H)	Magnetars, SCm critical-density regime

Implementation status: all 4 modes operational in `MAIN_1_CoAnQi.cpp`, `CondensedPhysics.py`, and `CondensedPhysics2.py`.